Physical Chemistry (I)

Lecture 24

기말고사 안내 및 복습

기말고사 안내

- 시간: 6월 13일(목) 오후 1:30~2:30
- 장소: 화학관 315호(홀수 학번), 112호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

- 8장 “단순 혼합물” 중 2차 중간고사 이후 범위
- 9장 “화학 평형”
- 10장 “전기화학”

시험 문제

총 여섯 문제

- 과제 문제 혹은 그 변형(2문제)
  - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 식 유도 및 응용(3문제)
  - 강의 자료 기준
- 보너스 문제(1문제)

Chapter 8

- Phase
- Chemical potential
- Gibbs-Duhem eq.

Multi-component system

Gas mixture

- Dalton's law
- Perfect gas
- Real gas

Ternary systems

Liquid mixture

Ideal solution

- Raoult's law

$$p_{\mathrm{A}} = x_{\mathrm{A}} p_{\mathrm{A}}^{*}$$

- Chemical potential

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{*} + RT \ln x_{\mathrm{A}}$$

- $\Delta_{\mathrm{mix}} G = -T \Delta_{\mathrm{mix}} S$

Ideal-dilute solution

- Henry's law

$$p_{\mathrm{B}} = x_{\mathrm{B}} K_{\mathrm{B}}$$

- Chemical potential

$$\mu_{\mathrm{B}} = \mu_{\mathrm{B}}^{\ominus} + RT \ln x_{\mathrm{B}}$$

Real solution

- Regular solution
- Bragg-Williams model
- Excess function
- Activity

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{*} + RT \ln a_{\mathrm{A}}$$

$$\mu_{\mathrm{B}} = \mu_{\mathrm{B}}^{\ominus} + RT \ln a_{\mathrm{B}}$$

- L-G phase diagram
- L-L phase diagram
- L-S phase diagram

Colligative properties

Concepts and Derivations in Chpt 8

- [C] Mixing of liquids: liquid-gas equilibrium
- [C] Ideal solutions
  - [C] Raoult's law and ideal solutions
  - [D] Chemical potential
  - [D] Thermodynamics of mixing
- [C] Ideal-dilute solutions
  - [C] Henry's law and ideal-dilute solutions
  - [C] (Idealized) chemical potentials of solute and solvent

Concepts and Derivations in Chpt 8

- [C] Regular solutions
- [C] Bragg-Williams model
  - [D] Entropy of mixing
  - [D] Energy of mixing
  - [D] Helmholtz energy of mixing
- [C, D] Excess function
- [C, D] Activity and activity coefficient
  - [D] Activity of Bragg-Williams model

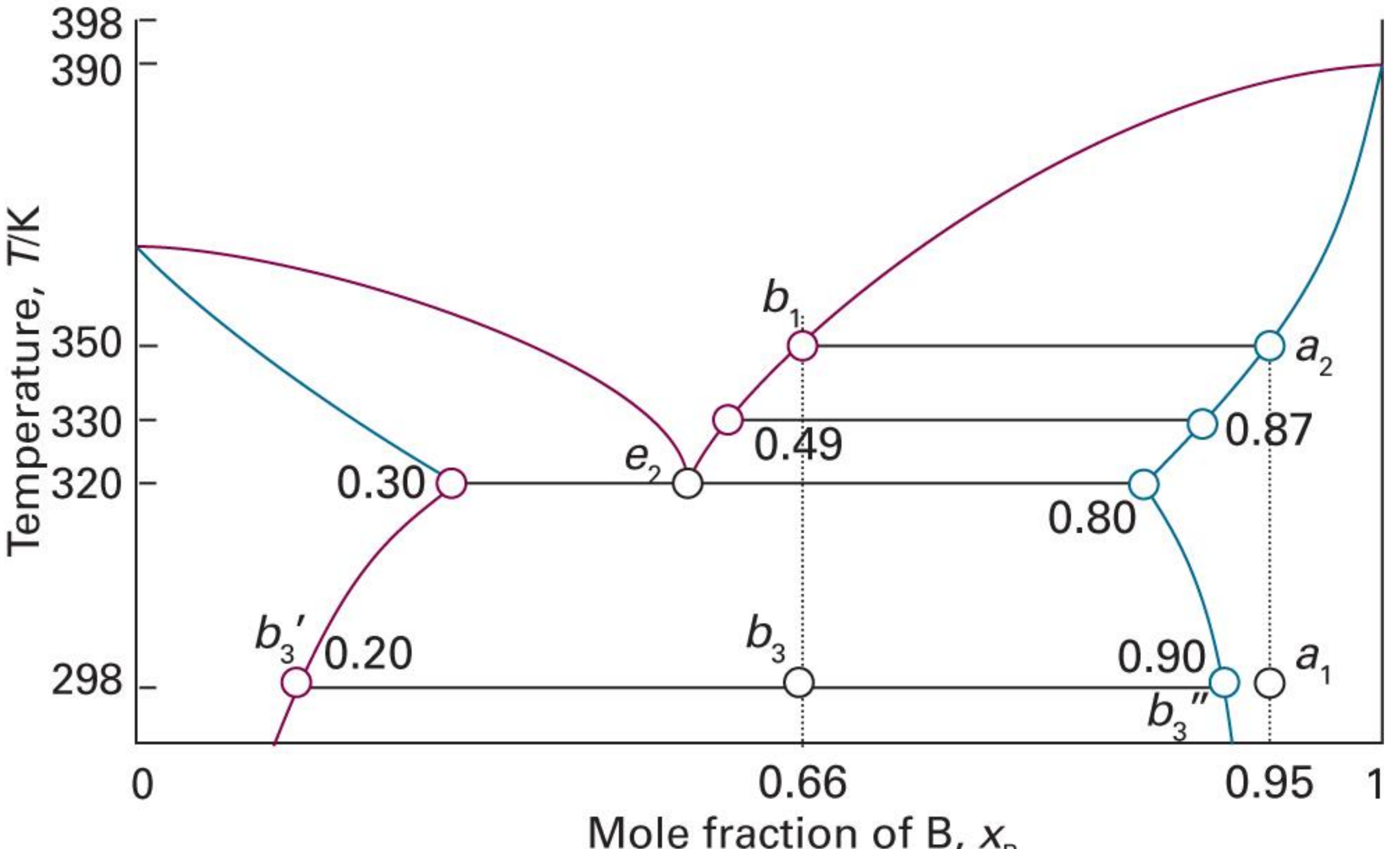
Concepts and Derivations in Chpt 8

- [C, D] Phase diagrams of ideal solutions
  - [D] Liquid and vapor compositions
  - [C] Interpretation of phase diagrams
  - [C, D] Lever rule
- [C, D] Phase diagrams of real solutions
  - [C] azeotropes
  - [C, D] phase separation
- [C] Phase diagrams of liquid-solid mixtures

Concepts and Derivations in Chpt 8

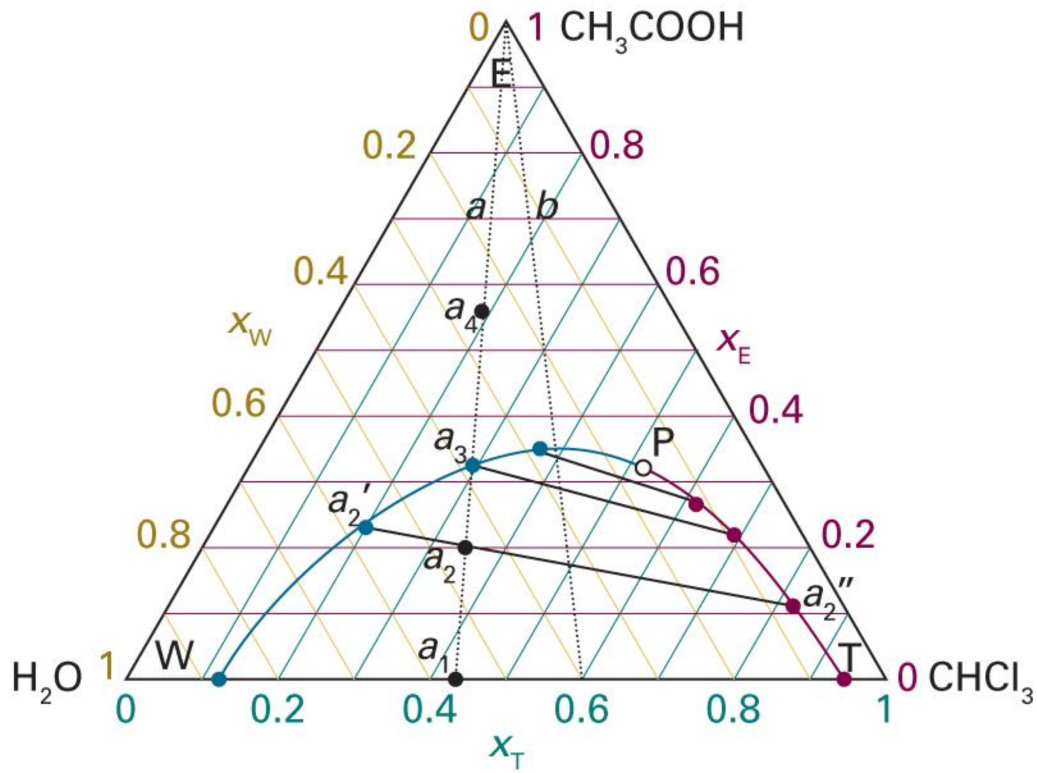
- [C, D] Colligative properties
  - Lowering of vapor pressure
  - Elevation of T_b and depression of T_f
  - Osmotic pressure
  - van’t Hoff factor
- [C] Phase diagrams of ternary systems

Interpretation of a Phase Diagram



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5C.23)

Interpretation of a Phase Diagram



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.4)

Chapter 9

Multi-component Gibbs energy

$$\mathrm{d}G = \sum_i \mu_i \mathrm{d}n_i$$

Chemical reaction

- Extent of reaction

$$\mathrm{d}n_i = \nu_i \mathrm{d}\xi$$

Gaseous reactions

- Equilibrium constants

K_p and K_c

- Theoretical calculation

- Reaction quotient
- Equilibrium constant

Le Chatelier's principle

- Amount change
- Volume change
- Temperature change
  - van't Hoff equation

Chemical potential

$$\mu = \mu^{\ominus} + RT \ln a$$

Activity approximations

Concepts and Derivations in Chpt 9

- [C, D] Reaction quotient and equilibrium constant
- [C, D] Activity approximations
- [D] Gaseous reactions
  - [D] K_p and K_c
  - [D] Calculation of K_c from partition functions
- [C, D] Le Chatelier's principle
  - Amount change
  - Volume change
  - Temperature change: van't Hoff equation

Chapter 10

Ions (electrolytes)

$$\delta w_{\mathrm{elec}} = \phi \mathrm{d}Q$$

- Poisson equation
- Boltzmann distribution

Solvation

Born equation

Electric potentials

Poisson-Boltzmann eq.

Debye-Hückel eq.

- Ion strength
- Debye radius

Activity

Debye-Hückel law

- Determination of $\gamma_{\pm}$

Electrochemical reaction

- Electrochemical potential

$$\mu_J' = z_J F\phi + \mu_J$$

$$\mathrm{d}G_{T,p} = \delta w_{\mathrm{add,max}}$$

Electrochemical cell

- Cell potential

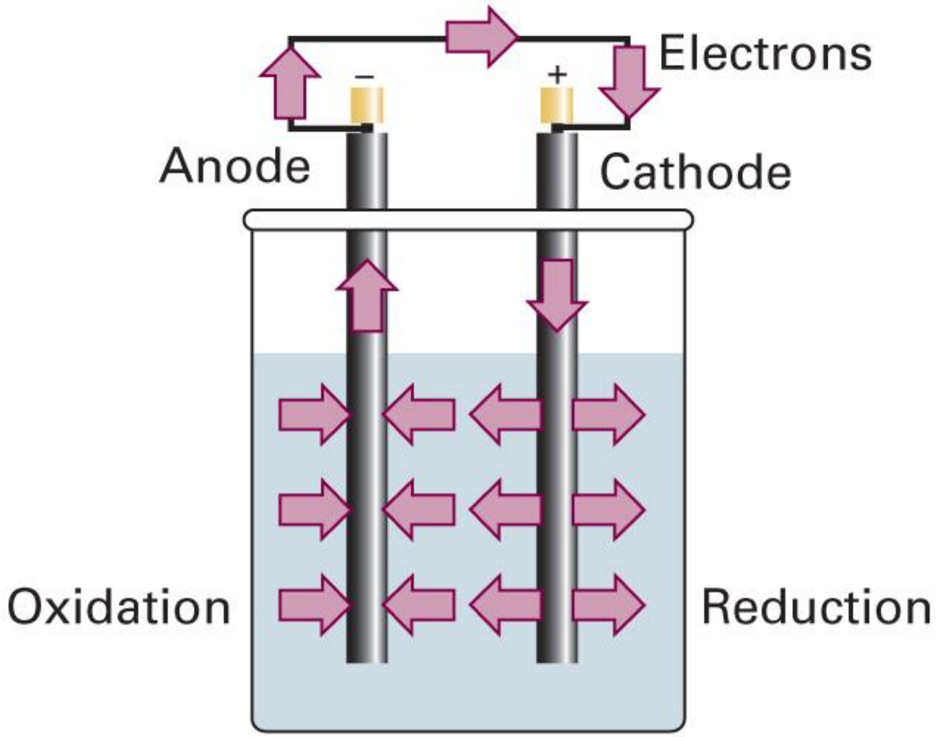
$$\Delta_{\mathrm{r}} G = -\nu F E_{\mathrm{cell}}$$

- Nernst equation

$$E_{\mathrm{cell}} = E_{\mathrm{cell}}^{\ominus} - \frac{RT}{\nu F}\ln Q$$

- Electrode potentials

Electrochemical Cell



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 6C.1)

Concepts and Derivations in Chpt 10

- [C, D] Gibbs energy of ion solvation: Born equation
- [D] Ion potential: Debye-Hückel equation
- [D] Ion activity: Debye-Hückel limiting law
- [C, D] Thermodynamics of non-expansion works
- [C, D] Electrochemical potentials
  - [D] Ion distributions
  - [D] Electrode-ion equilibrium

Concepts and Derivations in Chpt 10

- [C] Electrochemical cells
- [D] Thermodynamics of cells
- [D] Nernst equation
- [D] Entropy and enthalpy of cell reactions
- [C, D] Electrode potentials
- [D] Harned cell: determination of $E^{\ominus}$ and $\gamma_{\pm}$